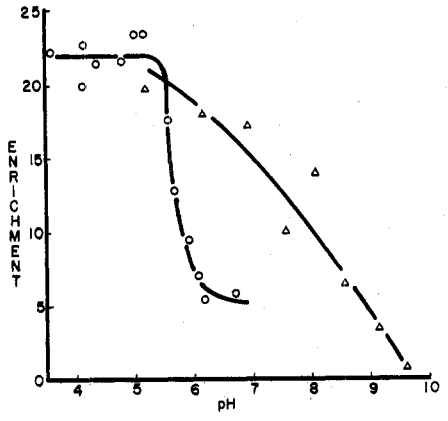


Figure 2. Influence of pH and sample composition on the enrichment of nickel: receiver electrolyte 1.0 M $MgSO_4$, 5.0×10^{-3} M $Al_2(SO_4)_3$; O, 10^{-3} M phosphate; Δ, 10^{-3} M carbonate; sample pH adjusted with HCl or NaOH; sample ionic strength 3×10^{-3} M adjusted with NaCl.

factors of both Ni(II) and Co(II) were found to vary widely. For example, enrichment factors obtained for Ni(II) at pH 8 were 1.7, 2.3, 8.1, and 13.4. Above pH 10 no enrichment of either Ni(II) or Co(II) was obtained. This is identical with the data obtained for the carbonate-containing system and is due to the formation of insoluble metal hydroxides.

The data in Figure 2 offers a means of eliminating problems due to both pH and phosphate in the sample by adjusting the pH of the sample to below 5. Adjusting the pH of the digested sample prior to measuring the resistance and enrichment resulted in improvements in both the enrichment factor and reproducibility. Enrichment factors obtained from pH-adjusted river water samples were in the range expected based on the data in Figure 1.

Linear working curves were obtained from the spiked samples for the determination of Ni in river water after UV digestion, pH adjustment, and Donnan dialysis enrichment. A least-squares curve fit of the data gave a slope of 5.16, intercept of 0.007, and a correlation coefficient of 0.999. This compares favorably with the data obtained for the enrichment of standard solutions containing only Ni and Co which gave a slope of 5.17 and a correlation coefficient of 0.999. Recoveries of Ni from spiked river water samples as determined from the standard solution working curve were 199 ± 5 ppb for a 200-ppb spiked sample and 6.0 ± 1.8 ppb for a 5-ppb spiked sample, each based on four replicates.

The close agreement between the working curves obtained for standard samples and spiked river water samples is an indication of the successful application of Donnan dialysis enrichment to the determination of trace metals in natural hard waters. The ability to obtain accurate quantitative results using a working curve prepared from pure aqueous standard solutions greatly simplifies the procedure. Since it is not necessary to match the sample matrix one standard working curve can be used for a wide variety of samples of varying composition.

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Theory of Square Wave Voltammetry for Kinetic Systems

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The theoretical response for the application of square wave voltammetry to systems complicated by electrode kinetics or by preceding, following, or catalytic homogeneous chemical reactions is presented. Experimentally measurable parameters such as peak shifts, heights, and widths are calculated and plotted as functions of the appropriate rate constants. These curves are characteristic of the electrode process and provide a basis for the extraction of kinetic information from the fast scan square wave experiment.

The theory and experimental verification for the application of a generalized square wave waveform to a reversible system have been presented (*1*, *2*). This technique has also been shown to give well-defined peaks at concentrations levels of 10^{-7} M (*3*), thus ranking it among the most sensitive modern electroanalytical techniques. We believe that square wave voltammetry by virtue of its immunity to charging currents can be utilized in kinetic studies of chemical systems at concentrations below those now typically employed with other techniques. Schaar and Smith have reported the estimation of kinetic parameters for a fast one-electron-transfer process at 8×10^{-5} M concentration (*4*) using ac polarography. However, loss of sensitivity for irreversible reactions is a much more serious problem in ac voltammetry than in square wave voltammetry, as will be illustrated below. It should be possible in square wave voltammetry to obtain sufficiently precise data

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Table I. Reaction Schemes and Their Rate Parameters

mechanism	reaction	parameters
1. slow electron transfer	$O + ne \overset{\kappa}{\rightleftarrows} R$	$\kappa = k_a^0/D_R^{(\alpha/2)}D_O^{(1-\alpha)/2}$
2. preceding chemical reaction	$Y \underset{k_b}{\overset{k_f}{\rightleftarrows}} O$	$k = k_f + k_b$
	$O + ne^- \rightleftarrows R$	$K = k_f/k_b$
3. following chemical reaction	$O + ne^- \rightleftarrows R$	$k = k_f + k_b$
	$R \underset{k_b}{\overset{k_f}{\rightleftarrows}} Y$	$K = k_f/k_b$
4. catalytic chemical reaction	$O + ne^- \rightleftarrows R$	k_c
	$R + Y \overset{k_c}{\rightleftarrows} O + Z$	

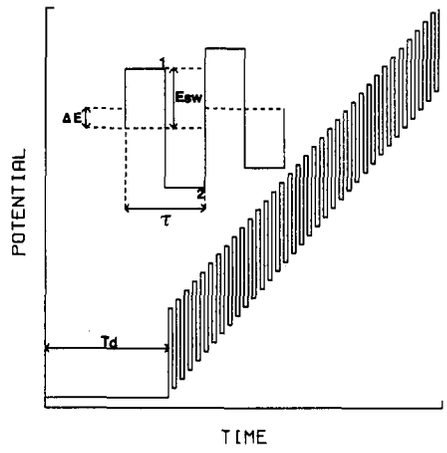


Figure 1. Square wave waveform showing the step height, ΔE, square wave amplitude, E_{sw}, square wave period, τ, delay time, T_d, and current measurement times, 1 and 2.

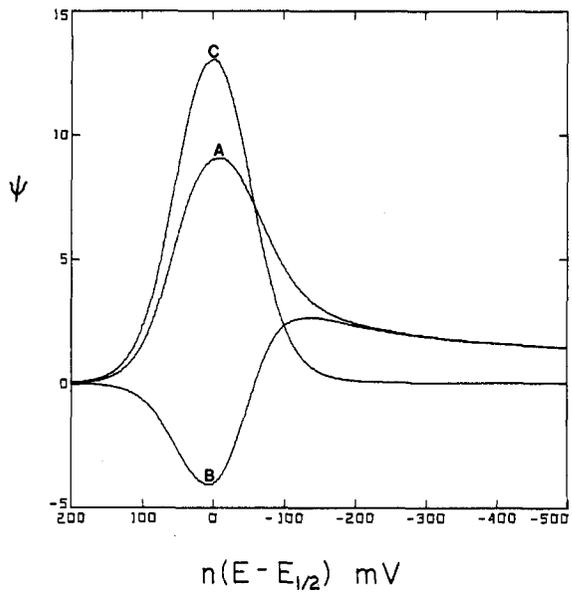


Figure 2. Calculated voltammograms for reversible electron transfer: (A) forward, (B) reverse, and (C) net current in dimensionless units (see eq A4).

to determine kinetic parameters accurately over the concentration range 10^{-5}–10^{-3} M and thus to use concentration as a parameter of the system being studied. Also, as square wave voltammetry finds broader application as an analytical technique, it is essential to understand how complications to a simple reversible electrode process affect the current response. This work extends the theory of square wave voltammetry to include electrochemical systems complicated by first-order homogeneous preceding, following, and catalytic chemical reactions and by slow electron transfer.

The excitation waveform of square wave voltammetry is shown in Figure 1. The nomenclature of Christie (*1*) has been adopted. Four parameters characterize the waveform. The step height of the base staircase is ΔE, E_{sw} is the amplitude of the square wave superimposed upon the staircase (note that E_{sw} is half the peak-to-peak amplitude), and τ is the square wave period, i.e., the step width of the base staircase. An additional parameter, T_d, specifies an arbitrary delay time before the wave form is applied. The waveform considered in this study is symmetrical. Forward currents are measured at t_1 just before the "down" pulse is applied. Reverse currents are measured at t_2. The forward difference or net current is defined as the current measured at t_1 minus the current measured at t_2. A plot of these currents calculated for a reversible process at a planar electrode is shown in Figure 2. Note that the abscissa is measured in units of $n(E - E_{1/2})$ mV, where n is the number of electrons transferred, and that the ordinate is measured in units of dimensionless current, ψ, as defined in eq A4. Unless otherwise specified, in the following current refers to the net current. For our calculations a $10n$ mV step height was selected. This value of ΔE is small enough to define peak shape well and yet is large enough to permit a moderate scan rate, i.e., $10n$ V s^{-1} at 1000 Hz. The square wave amplitude was chosen such that nE_{sw} = 50 mV. The ratio of peak current (ψ_p) to peak width ($nW_{1/2}$, mV) exhibits a maximum at nE_{sw} = 50 mV for a reversible electron transfer. Figure 3 shows that this maximum is broad and largely independent of the step height chosen for the waveform. Clearly step heights as large as $25n$ mV could be employed with little loss in analytical response if experimental constraints demanded even faster scan rates.

The step function method elaborated by Nicholson and Olmstead (*5*) was used to solve the integral equations which describe each of the four systems listed in Table I. The integral equations for the preceding, following, and catalytic cases at a planar electrode have been derived by Smith (*6*) in connection with ac polarography and form the basis of the solutions presented here. The actual expressions obtained are general solutions for the response of an electrochemical system to an arbitrary excitation waveform composed of steps. Hence, the application of these solutions is limited only by the ingenuity of the investigator in composing a waveform which will aid in the elucidation of a particular system under study. Of course, the "arbitrary" waveform employed in this study is the square wave waveform. Similar calculations treating the waveforms of reverse pulse (*7*), differential normal pulse (*8*), and staircase (*9*) voltammetry are in progress. All computer programs were coded in standard Fortran IV and are available upon request. Details of the method of calcu-

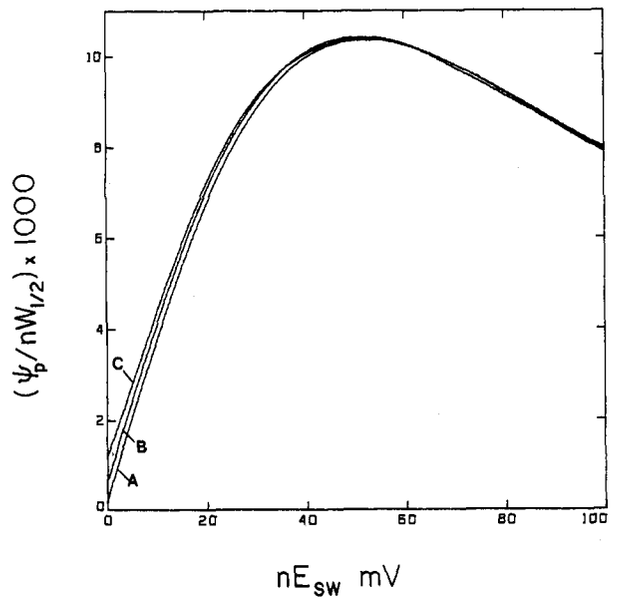


Figure 3. Peak height (ψ_p)/peak width ($nW_{1/2}$, mV) vs. E_{sw} for three values of ΔE: (A) $\Delta E = 2.5n$ mV, (B) $\Delta E = 10n$ mV, (C) $\Delta E = 25n$ mV.

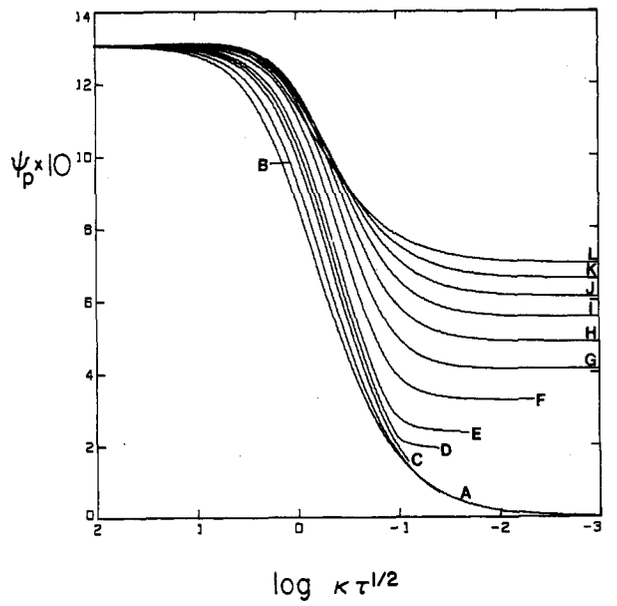


Figure 5. Peak height (ψ_p) for electrode kinetics as a function of log ($\kappa\tau^{1/2}$) and several values of α: (A–L) α = 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0.

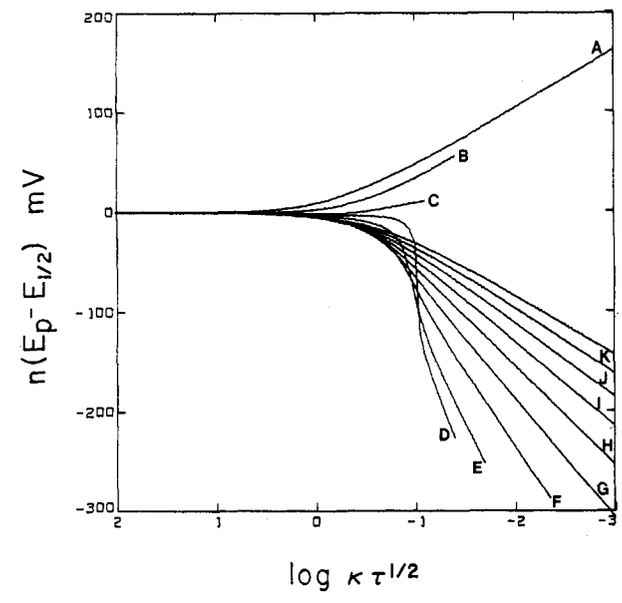


Figure 4. Peak position, $n(E_p - E_{1/2})$, for electrode kinetics as a function of log ($\kappa\tau^{1/2}$) for several values of α: (A–L) α = 0, 0.1, 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0.

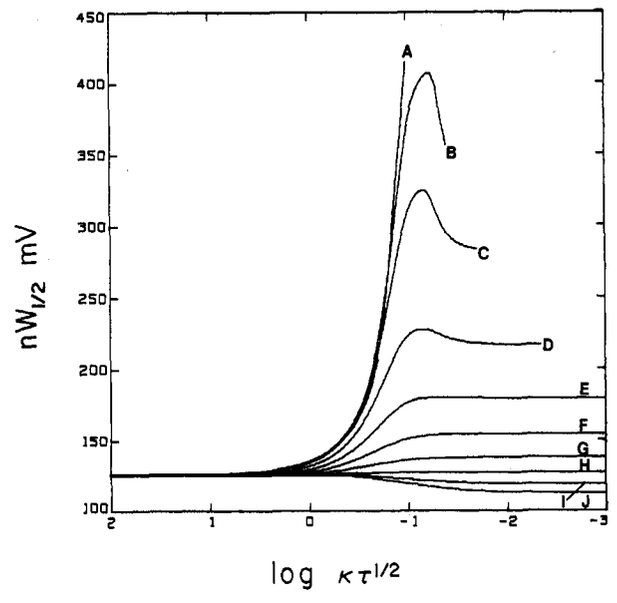


Figure 6. Peak width ($nW_{1/2}$, mV) for electrode kinetics as a function of log ($\kappa\tau^{1/2}$) and several values of α: (A–J) α = 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0.

lation are given in the Appendix.

CASE 1. SLOW ELECTRON TRANSFER

Figure 4 shows how the position of the peak, E_p, arising from a slow electron transfer process varies as a function of log ($\kappa\tau^{1/2}$) for several values of α (cf. Table I). Figures 5 and 6 show the peak height, ψ_p, and width, $W_{1/2}$, respectively, over the same range of log ($\kappa\tau^{1/2}$). Three distinct types of behavior are observed. First, for log ($\kappa\tau^{1/2}$) > 1, the reaction behaves reversibly. The peak position is coincident with the theoretical half-wave potential and the peak height and width are independent of $\kappa\tau^{1/2}$. Second, there is a transitional region, 1 > log ($\kappa\tau^{1/2}$) > −1, where quasi-reversible behavior is seen. The peak position, height, and width all vary with $\kappa\tau^{1/2}$. Finally, for log ($\kappa\tau^{1/2}$) < −1 the behavior is totally irreversible and the peak height and width are again independent of $\kappa\tau^{1/2}$.

The quasi-reversible region is of some interest since dramatic changes in peak shape occur within it. The situation becomes especially complicated for values of α less than 0.5. Compare the peak position, height, and width for α equal to 0.25 with the same calculation for α equal to 0.5 as shown in Figures 4–6. The most notable features of the curves for α = 0.25 are the abrupt change of peak position and the hump in peak width that occur at log ($\kappa\tau^{1/2}$) ≈ 0.1. These are caused by the occurrence of dual peaks in the voltammograms as shown in Figure 7. The appearance of these peaks is due to a reverse current response which is skewed in the positive direction from the forward current response. The result as seen in the net current is the impression of overlapping peaks. Similar effects have been predicted and verified for ac polarography (*10*) and subsequently for differential pulse polarography (*11*). For sufficiently small values of α (<0.2), the skewing of forward and reverse currents is so severe that the peak position actually shifts toward more positive potentials, and the peak exhibits a large, drawn-out tail on the negative side. This effect is shown in Figure 8.

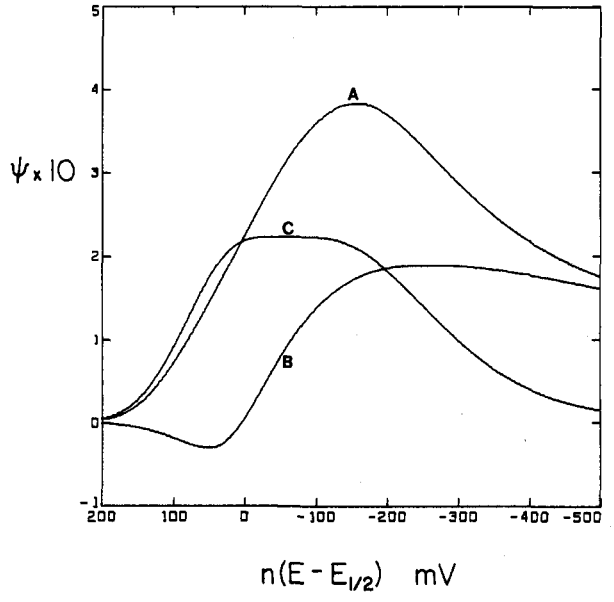


Figure 7. Voltammograms for electrode kinetics, α = 0.25: (A) forward, (B) reverse, and (C) net current in dimensionless units.

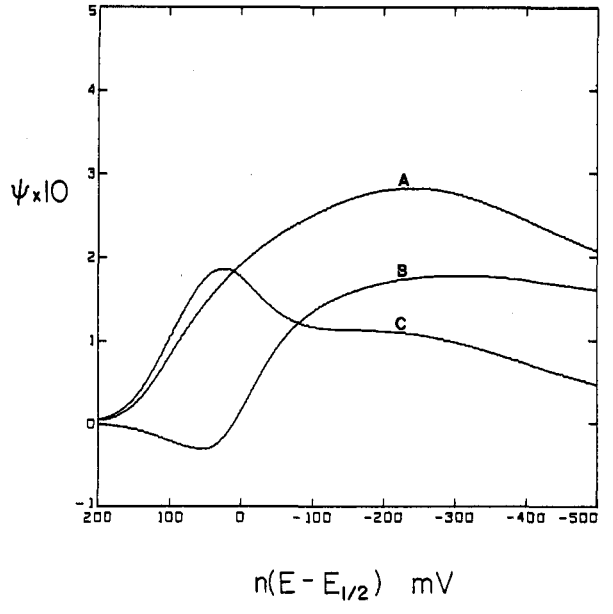


Figure 8. Voltammograms for electrode kinetics, α = 0.15: (A) forward, (B) reverse, and (C) net current in dimensionless units.

CASE 2. PRECEDING CHEMICAL REACTION

Calculations for a chemical reaction preceding electron transfer were carried out for several values of log K, where K is the equilibrium constant (cf. Table I). Figures 9–11 show how the peak position, height, and width vary as a function of log ($k\tau$) where k is the sum of the forward and reverse first-order homogeneous rate constants. As expected, as the square wave period, τ, becomes longer and longer more material has time to convert to an electroactive form and the peak current increases. As with cyclic voltammetry (*12*) and ac polarography (*13*) the peak position shifts in the negative direction as the rate of the preceding reaction increases. The peak width is largely insensitive to the preceding reaction and varies no more than −5% from the reversible value of 126n mV.

CASE 3. FOLLOWING CHEMICAL REACTION

Calculations for a reversible electron transfer followed by a chemical reaction were also carried out for several values of log K (cf. Table I). Figures 12–14 show the dependence

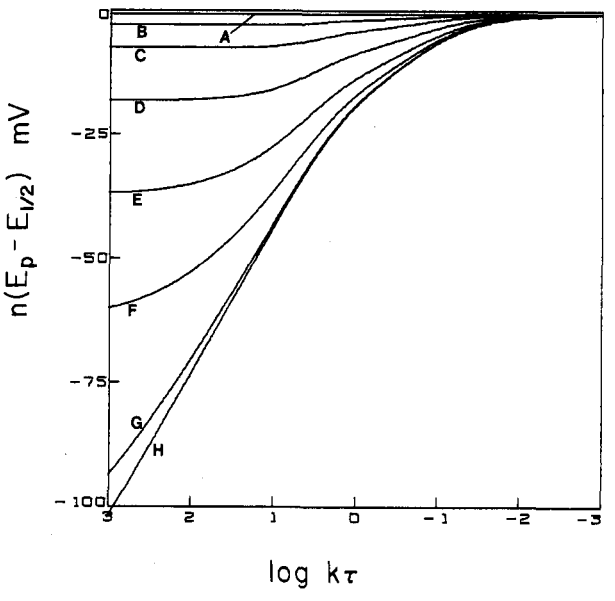


Figure 9. Peak positions, $n(E_p - E_{1/2})$, for preceding reaction as a function of log ($k\tau$) for several values of log K: (A–H) log K = 2, 1, 0.5, 0, −0.5, −1, −2, and −3.

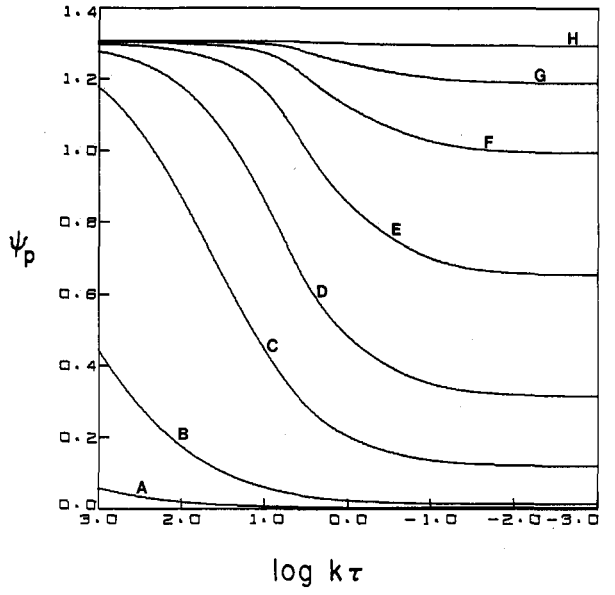


Figure 10. Peak height (ψ_p) for preceding reaction as a function of log ($k\tau$) for several values of log K: (A–H) log K = 2, 1, 0.5, 0, −0.5, −1, −2, and −3.

of peak position, height, and width upon log ($k\tau$). Again the peak potential is shifted to more positive values in harmony with the effect noted in cyclic voltammetry and ac polarography. For large values of $k\tau$, shifts have a limiting slope of 29.6/n mV per decade. As expected the peak current drops off to a constant value as the reaction becomes faster and faster. This decrease is due to the increased consumption of the reduced species by the chemical reaction, thereby making it unavailable for reoxidation during the experiment. Consequently a substantial fraction of the reverse current is unrealized and does not contribute to the net current. The peak width is not a strong function of log ($k\tau$) but rather is nearly constant as in the case of the preceding reaction.

CASE 4. CATALYTIC CHEMICAL REACTION

Calculations for the catalytic case show that the peak potential is coincident with that for the reversible case. The peak widths are similarly constant and equal to their corresponding reversible values. Figure 15 shows the rapid increase in peak

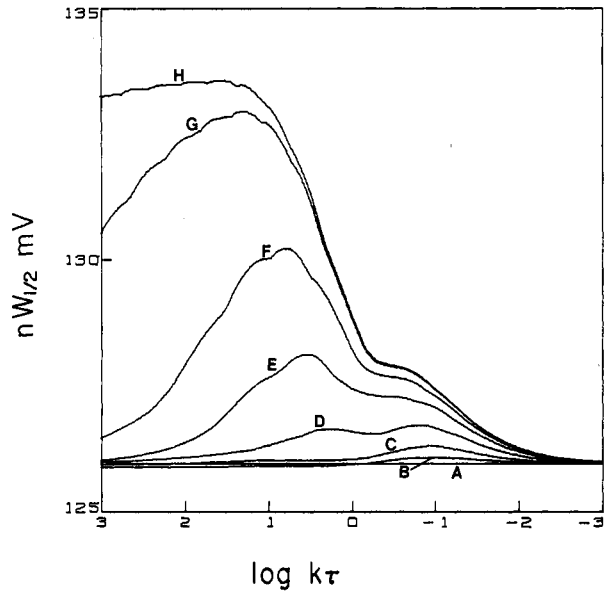


Figure 11. Peak widths, $nW_{1/2}$, for preceding reaction as a function of log ($k\tau$) for several values of log K: (A–H) log K = 2, 1, 0.5, 0, 0.5, −1, −2, and −3.

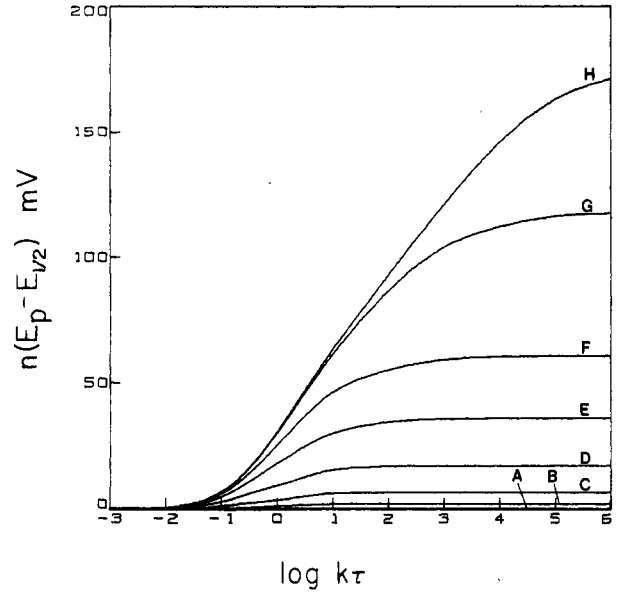


Figure 12. Peak position, $n(E_p - E_{1/2})$, for following reaction as a function of log ($k\tau$) for several values of log K: (A–H) log K = −2, −1, −0.5, 0, 0.5, 1, 2, and 3.

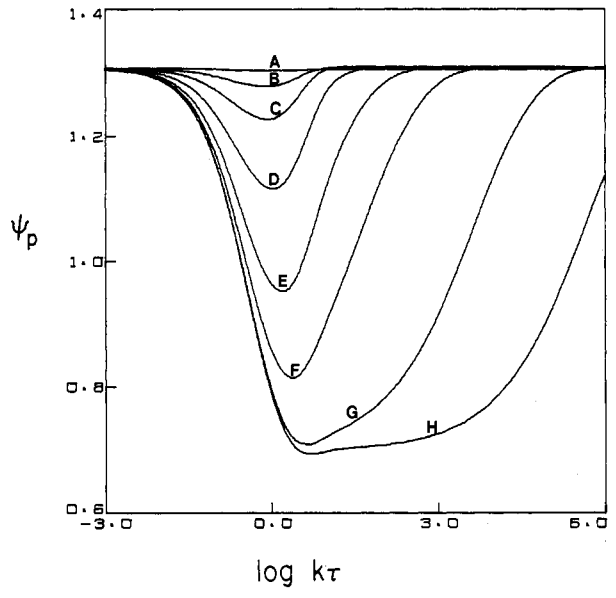


Figure 13. Peak height (ψ_p) for following reaction as a function of log ($k\tau$) for several values of log K: (A–H) log K = −2, −1, −0.5, 0, 0.5, 1, 2, and 3.

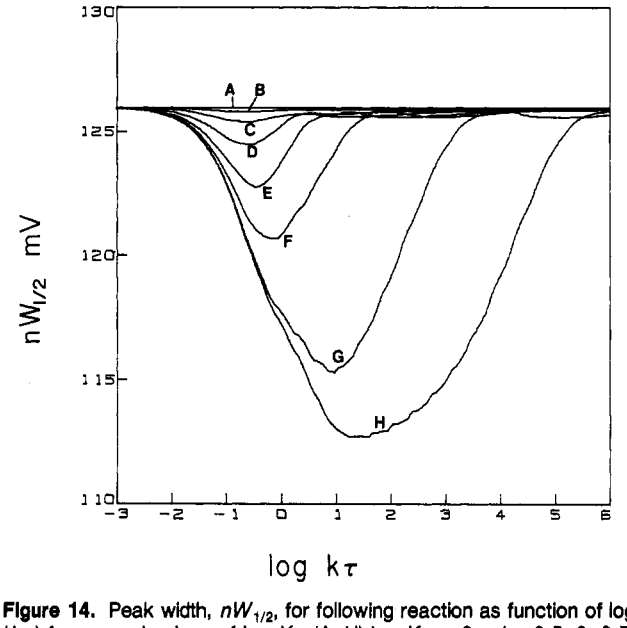


Figure 14. Peak width, $nW_{1/2}$, for following reaction as function of log ($k\tau$) for several values of log K: (A–H) log K = −2, −1, −0.5, 0, 0.5, 1, 2, and 3.

current as the catalytic reaction comes into play. This current increases without limit as the catalytic rate constant k_c, increases. At large values of k_c the electrode surface is supplied with a high flux of reducible material due to the continuous chemical reoxidation of the species discharged.

The catalytic reaction has a profound influence upon the individual forward and reverse currents long before its presence is manifested in the net current. Figure 16 shows this effect clearly. In the first instance, curve A ($k_c\tau$ = 0.15), the forward and reverse currents measured near the end of the scan ($\approx -300n$ mV) are over twice as large as the corresponding currents seen in curve D ($k_c\tau$ = 0.0015), which exhibits nearly reversible behavior. The corresponding net current curves B and C, however, differ by less than 4% in peak height and are nearly superimposible. For larger values of $k_c\tau$ the forward and reverse currents adopt a characteristic sigmoid shape.

Detailed experimental verification of these calculations will appear in a later publication. Furthermore, the extension of the mathematical approach described here to more complicated cases should be possible. Because accurate data can be obtained at low concentrations, it should have special use in the study of fast reactions.

APPENDIX

Slow Electron Transfer. We shall first consider the case of slow electron transfer:

$$O + ne^- \overset{\kappa}{=} R \qquad \text{(A1)}$$

The integral equations describing the surface concentrations of the species for planar semiinfinite boundary conditions are

$$C_O(0,t) = C_O^* - (C_O^*/\pi\sqrt{\tau}) \int_0^t (\psi(u)/\sqrt{t-u})\, du \qquad \text{(A2)}$$

$$C_R(0,t) = (\sqrt{D_O}C_O^*/\pi\sqrt{D_R\tau}) \int_0^t (\psi(u)/\sqrt{t-u})\, du \qquad \text{(A3)}$$

where $\psi(t)$ is a dimensionless current function given by

$$i(t) = \psi(t)\, nFA\, \sqrt{D_O}C_O^*/\sqrt{\pi\tau} \qquad \text{(A4)}$$

The Erdy-Gruz and Volmer equation is

$$i(t) = (nFAk_a^0/D_O^{(1-\alpha)/2}D_R^{(\alpha/2)})\exp(-(\alpha nF/RT) \times (E - E_{1/2}))[\sqrt{D_O}C_O(0,t) - \epsilon\sqrt{D_R}C_R(0,t)] \qquad \text{(A5)}$$

For convenience we define

$$\kappa = k_a^0/D_O^{(1-\alpha)/2}D_R^{\alpha/2} \qquad \text{(A6)}$$

$$\epsilon = \exp[(nF/RT)(E - E_{1/2})] \qquad \text{(A7)}$$

Combination of the above equations yields a new integral equation

$$\psi(t) = \kappa\epsilon^{-\alpha}\left(\sqrt{\pi\tau} - [(1 + \epsilon)/\sqrt{\pi}]\int_0^t [\psi(u)/\sqrt{t - u}]\, du\right) \qquad \text{(A8)}$$

By the method of Nicholson and Olmstead (5) we now replace the integral in eq A8 with a finite sum and $\psi(t)$ by $b(m)$, the approximation to $\psi(t)$. Further algebraic manipulations eventually yield the explicit solution for $b(m)$

$$b(m) = \left(\pi\sqrt{l}/\sqrt{2}(1 + \epsilon) - \sum_{i=1}^{m-1} b_i S_j'\right)\Big/\left(1 + \sqrt{\pi l}/\kappa\sqrt{2\tau}\epsilon^{-\alpha}(1 + \epsilon)\right) \qquad \text{(A9)}$$

Here l is the number of subintervals per half-period, b_i is the approximation to $\psi(t)$ at $t = m\tau/2l$, $S_j' = \sqrt{j} - \sqrt{j-1}$ and $j = m - i + 1$.

Preceding Chemical Reaction. Equation A10 shows the case of reversible electron transfer preceded by a chemical reaction

$$Y \underset{k_b}{\overset{k_f}{\rightleftharpoons}} O \qquad O + ne^- = R \qquad \text{(A10)}$$

The integral equations (A11) and A12) are taken from Smith (6)

$$C_O(0,t) = C_O^* - (1/(1 + K)nFAD_O^{1/2})\int_0^t (e^{-k(t-u)}i(u)/\sqrt{\pi}\sqrt{t - u})du - (K/(1 + K)nFAD_O^{1/2})\int_0^t (i(u)/\sqrt{\pi}\sqrt{t - u})\, du \qquad \text{(A11)}$$

and

$$C_R(0,t) = (1/nFA\sqrt{D_R})\int_0^t (i(u)/\sqrt{\pi}\sqrt{t - u})\, du \qquad \text{(A12)}$$

where

$$K = k_f/k_b \qquad \text{(A13)}$$

$$k = k_f + k_b \qquad \text{(A14)}$$

and it is assumed that $D_Y = D_O$. Since O and Y are in equilibrium before the experiment

$$C_O^* = KC_Y^* \qquad \text{(A15)}$$

The total analytical concentration of the reactant C^* is

$$C^* = C_O^* + C_Y^* = C_O^*(1 + 1/K) \qquad \text{(A16)}$$

For this case C^* replaces C_O^* in eq A4. Again the current function can be substituted into the integral equations for the surface concentrations $C_O(0,t)$ and $C_R(0,t)$. Since the electron-transfer process is reversible, eq A11 and A12 are substituted into the Nernst equation to give

$$(1/(K + 1))\int_0^t (e^{-k(t-u)}\psi(u)/\sqrt{t - u})\, du = K\pi\sqrt{\tau}/(K + 1) - (\epsilon + K/(K + 1))\int_0^t (\psi(u)/\sqrt{t - u})\, du \qquad \text{(A17)}$$

Equation A17 is solved as suggested above to yield an equation for $b(m)$

$$b(m) = \left[K\pi\sqrt{k\tau}/(K + 1) - \left(\sqrt{\pi}/(K + 1)\right)\sum_{i=1}^{m-1} b_i R_j' - \sqrt{2k\tau/l}(K/(K + 1) + \epsilon)\sum_{i=1}^{m-1} b_i S_j'\right]\Big/[\sqrt{\pi}R_1'(K + 1) + \sqrt{2k\tau/l}(K/(K + 1) + \epsilon)] \qquad \text{(A18)}$$

Here $R_j' = erf\sqrt{jk\tau/2l} - erf\sqrt{(j-1)k\tau/2l}$.

Following Chemical Reaction. The case of a reversible electron transfer followed by a chemical reaction is given by eq A19.

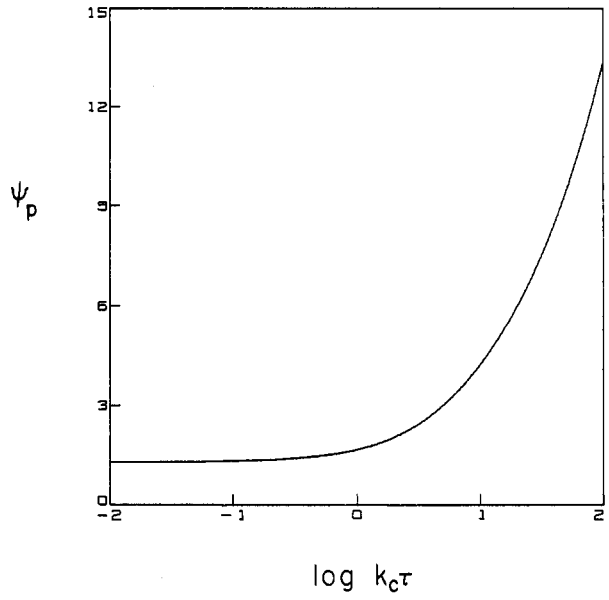


Figure 15. Peak height (ψ_p) as a function of log ($k_c\tau$) for catalytic reaction.

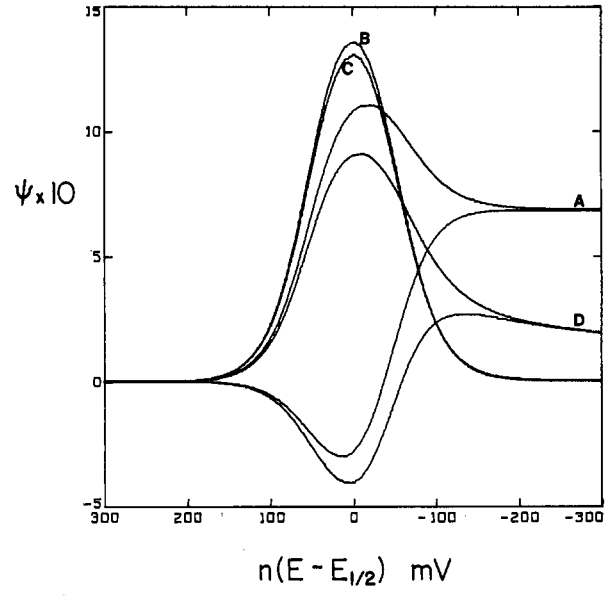


Figure 16. Voltammograms for catalytic reaction: (A) $k\tau = 0.15$, forward and reverse currents, (B) $k\tau = 0.15$, net current, (C) $k\tau = 0.0015$ net current, (D) $k\tau = 0.0015$, forward and reverse currents.

$$O + ne^- = R \qquad R \underset{k_b}{\overset{k_f}{\rightleftarrows}} Y \tag{A19}$$

The integral equations from Smith (5) are

$$C_O(0,t) = C_O^* - (1/nFA\sqrt{\pi D_O}) \int_0^t (i(u)/\sqrt{t-u})\, du \tag{A20}$$

and

$$C_R(0,t) = (1/(1+K)nFA\sqrt{\pi D_R}) \int_0^t (i(u)/\sqrt{t-u})\, du + (K/(1+K)nFA\sqrt{\pi D_R}) \int_0^t (e^{-k(t-u)} i(u)/\sqrt{t-u})\, du \tag{A21}$$

in which it is assumed $D_Y = D_R$. Replacing the integrals by finite sums and introducing the Nernst equation gives an expression for $b(m)$

$$b(m) = \left[\sqrt{\pi k\tau}/\epsilon - (\sqrt{\pi}K/(1+K)) \sum_{i=1}^{m-1} b_i R_j' - \sqrt{2k\tau/l}(1/(1+K) + 1/\epsilon) \sum_{i=1}^{m-1} b_i S_j'\right] \Big/ \left[KR_1'\sqrt{\pi}/(1+K) + \sqrt{2k\tau/l}(1/(1+K) + 1/\epsilon)\right] \tag{A22}$$

Catalytic Reaction. The final case which will be treated is reversible electron transfer with an irreversible catalytic oxidation

$$O + ne^- = R$$

$$R + Y \xrightarrow{k_c} O + Z$$

The integral equations from Smith are

$$C_O(0,t) = C_O^* - (1/nFA\sqrt{\pi D}) \int_0^t (\exp(-k_c(t-u)) i(u)/\sqrt{t-u})\, du \tag{A23}$$

and

$$C_R(0,t) = (1/nFA\sqrt{\pi D}) \int_0^t (\exp(-k_c(t-u)) i(u)/\sqrt{t-u})\, du \tag{A24}$$

where $D = D_O = D_R$ and $C_Y^* \gg C_O^*$. Following the scheme outlined above, the approximation to $\psi(t)$ is obtained

$$b(m) = \left[\sqrt{\pi k_c\tau}/(1+\epsilon) - \sum_{i=1}^{m-1} b_i R_j'\right] \Big/ R_1' \tag{A25}$$

The solutions for the case of slow electron transfer complicated by homogeneous first-order reactions can be obtained by substituting integral equation pairs (A11) and (A12), (A20) and (A21), and (A23) and (A24) into the Erdy-Gruz and Volmer equation (eq A5) instead of the Nernst equation and solving for corresponding expressions for $b(m)$.

Equations A9, A18, A22, and A25 were coded into a Fortran IV program and were used to calculate 101 voltammograms as a function of the parameter of interest with all other parameters held constant. The resulting data sets consisted of discrete forward and reverse currents at $10n$ mV intervals. These were differenced to yield the corresponding net current voltammograms. The calculated net currents were analyzed as follows. The top most three points of the data set determine a parabola; the position and height of the vertex of this parabola were taken as the position and height of the peak. The peak width at half-height was taken as the length of a horizontal line segment at half-height bounded on both sides by its intersection with tangents to the peak. the peak position, height, and width were then plotted as a function of the parameters of interest to provide a visualization of the behavior of the system.

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